



DEALER BEST PRACTICE SERIES

Parts Buffer Enhancement

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1.0 Introduction

In a CRC, parts cleaning is typically performed in dedicated general parts cleaning work bay. Parts cleaning may utilize several technologies, including high-pressure cabinet washing, manual solvent washing, or gasket scraping. Parts cleaning must be managed to provide high quality rebuilds while minimizing costs and “over-cleaning”.

Clean parts introduce minimal contaminants into a component, reducing internal parts wear and providing longer service life. Clean parts also help assure that joints fit accurately and securely without future leaks. Worn parts often retain gasket material, hard carbon, and rust in their joint mating areas, even after high-pressure cabinet washing. These surface areas minimally require scraping and possibly buffing/blasting if the contamination is aggressive.

Many cleaning processes create airborne debris. Minimal effects find the debris deposited on staged clean parts and inside equipment, while a worst case could send debris through the entire shop rebuild/assembly areas. In all instances, uncontrolled airborne debris creates an adverse situation for repair quality.

This Best Practice describes a well-designed method of controlling the airborne debris generated when buffing parts. It uses a downdraft table, often considered a standard piece of CRC cleaning equipment, to help control buffing debris. These benches are typically used only with hand-held angle grinder type buffers but can be used more effectively with bench grinder type buffer.

2.0 Best Practice Description

Parts buffing is often performed with a handheld angle grinder over a downdraft table. A bench grinder buffer is typically a more efficient (faster) tool if the piece part can be conveniently hand-held. Controlling the bench grinder’s airborne debris is often difficult since the wheel guard extraction systems are somewhat ineffective.

The most effective debris extraction system utilizes a bench grinder mounted on a fabricated platform. The platform allows the buffer to be positioned into the downdraft table’s working area, where it is pinned in place. Once in-place, the platform assembly activates a switch allowing the operator to start the buffer. When the operator is finished with the buffer it can be stopped, unpinned, and pivoted outside the downdraft table working area to the side of the table. The table is now clean and available for standard use.

Downdraft tables are effective at controlling buffing debris. Compared to a wheel guard extraction system, the table draws much more air through the buffing wheels and guards. This arrangement catches much more buffing debris, preventing it from drifting onto the cleaning technicians, previously cleaned parts, or into the shop area.

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3.0 Implementation Steps

1. Determine area in shop to be designated for cleaning/buffing.
 - a. Assure cleaning area has supporting utilities (compressed air for angle buffer, lighting, and electricity as needed)
 - b. Assure area has adequate space and access for equipment and parts staging.
2. Design and construct enhanced buffing equipment
3. Size to needs
4. Establish equipment usability/configuration to assure easy cleaning/maintenance and professional appearance.
 - a. Establish new/adjusted repair/rebuild time requirement targets as needed.
5. Eliminate the current processes/tooling/equipment strategy (only one best way).
 - a. Install the layout/equipment changes into the shop
6. This may occur in phases, as needed, such as shop arrangement, then equipment final design/fabrication.
7. Promote change feedback as installation may surface new problems.
 - a. Train employees to use the new procedures/equipment. Follow-up/enforce immediately.
 - b. Establish equipment maintenance schedule/responsibility.
 - c. Review process/equipment performance once established.
 - d. Establish and install any adjustments as needed.

4.0 Benefits

Saves Labor – Well designed/organized cleaning equipment reduces buffing efforts. Product staged in accessible carts reduces space needed. Less effort is required to maintain a clean work area and fewer parts need follow-up cleaning before assembly.

Saves Time – Parts are not staged in different areas for different types of cleaning, reducing work flow interruptions. Protecting parts against airborne debris interferes less with cleaning.

Increases Quality – More effective cleaning equipment simplifies cleaning processes and promotes high technician professionalism.

5.0 Resources Required

- **Facility**
 - Re-organize workstation to allow for downdraft table and buffer placement, incoming and outgoing staging, and table maintenance as needed.
- **Support Equipment**
 - A bench type buffer and an effectively sized downdraft table with side and overhead panels.
 - A fabricated platform for the buffer. This would be attached to a pivot shaft.
 - A pivot shaft attached to the platform and suspended by flange-mounted bearings.

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- Fabricated plates (3).
- Two plates attached to the side of the downdraft table, serving as platforms for the flange-mounted bearings.
- One plate attached to the side of the downdraft table, servicing as a buffer platform pinning anchor.
- Flange mounted bearings (2)
- A fabricated pivot-locking pin with a retention chain.
- A micro-switch installed to assure the buffer cannot be started until the buffer is pivoted/locked in the proper position.
- An electric terminal box for safe connections and plug-ins.
- Electrical cords as needed between switches, terminals, and power supplies.
- Clip fasteners for electric cord routing/retention.
- Miscellaneous fastening hardware

- **People**

- Establish current cleaning concerns with team
- Establish process, layout, and equipment improvements through team.

- **Training**

- Define/establish/enforce new processes, roles and responsibilities with production teams.
- Remove replaced items (buffers/tables) from operations to prevent less effective cleaning.

6.0 Supporting Attachments / References

Several supporting attachments.

7.0 Related Best Practices

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Parts Blasting Enhancement

8.0 Acknowledgements

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